

Y branch connector



Nowadays 210mm cell sized panels are getting more and more popular on the solar market. The Max. output current is also trending to be larger. Thus SolaX has released fully compatible solar inverters to match the large input current of the latest panels for different solar power plants. Please refer to the compatible inverter list [here](#).

This article will introduce the Y branch connector from four aspects: background, introduction, application scenarios and associated configuration.

The connection of PV panels is an important part for using an inverter, the choice of panels and the way they are connected needs to meet certain requirements and an incorrect connection method may cause a fault.

For more details of designing a solar power plant, please check the article [here](#).

Background

Among the requirements for the connection of PV strings, the following focuses on current: Assuming that the Maximum Power Current and the Short-circuit Current of the PV strings are I_{mp_str} and I_{sc_str} , for an inverter, the Max. short circuit current and Max. input current are I_{mp_inv} and I_{sc_inv} ,

The PV strings that can be connected to the inverter need to meet the following requirements:

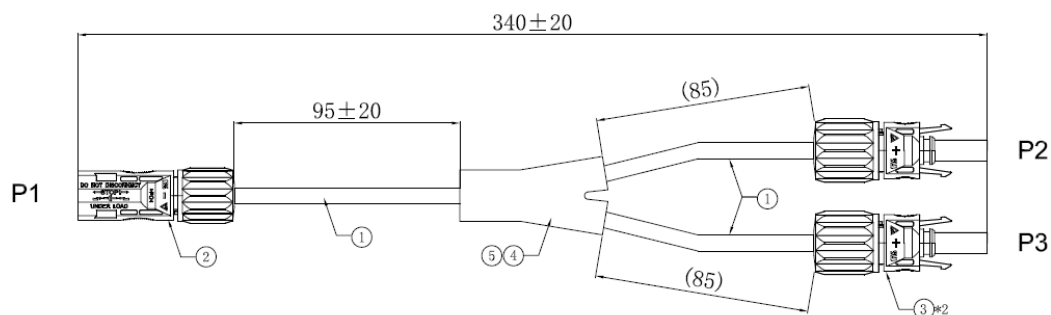
$$I_{mp_str} < I_{mp_inv}$$

$$I_{sc_str} < I_{sc_inv}$$

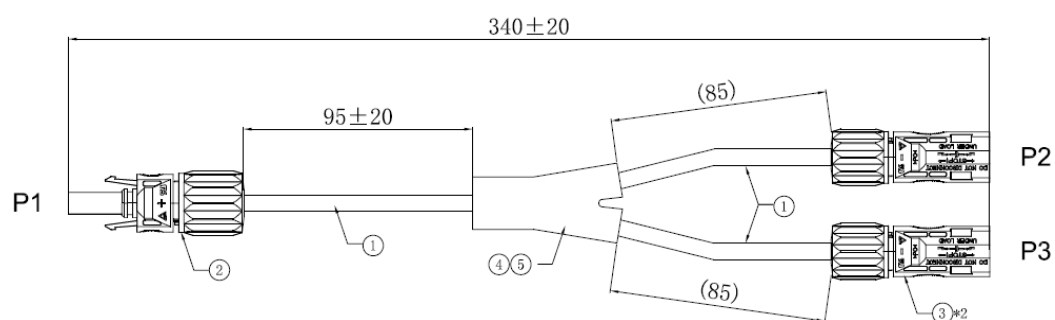
As can be seen by these limitations above, one of the key factors for a panel to work with an inverter is the current. In recent years, due to the upgrading of PV modules on the market, the current has gradually increased and exceeded the capacity of the input current allowed by the inverter. Thus, we introduce the Y- branch connector.

Introduction of the connector

The Y-branch connector is a forked connector with 2 terminals on one side and 1 on the other. The two ends of the connector is bi-directional, either end can be input, while the other end will be output, shown as below:



(1 Male and 2 Female Type Connector)



(1 Female and 2 Male Type Connector)

*Unit: mm

PV wire: 4mm² 2PFG 1169 PV1-F

Cable outside diameter: 6.0mm black

As for the using method, in addition to the connection to the PV panel and the inverter, the corresponding configuration on the inverter via the LCD screen is also required.

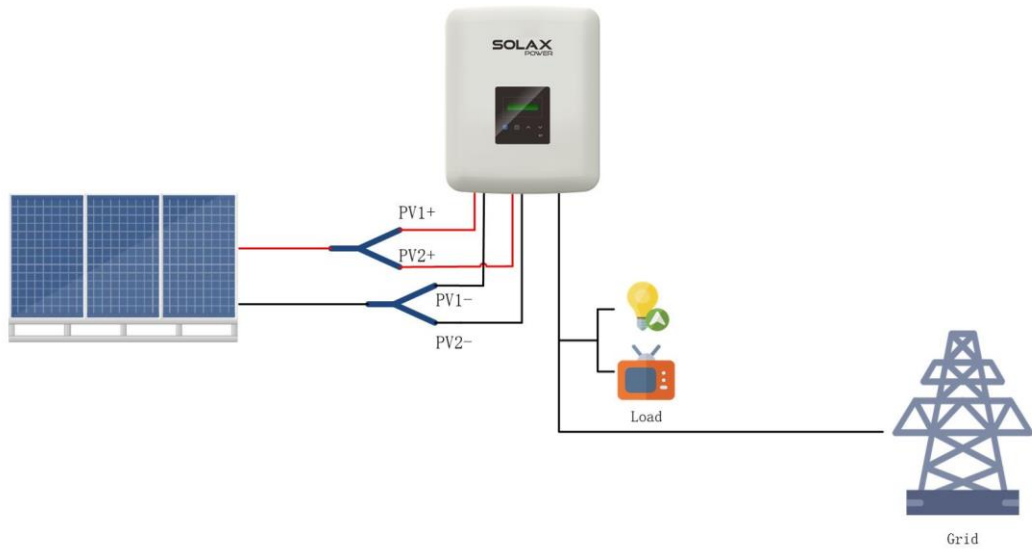
Application Scenarios

Y branch connector is recommended in the following scenarios:

High-current PV panel applications

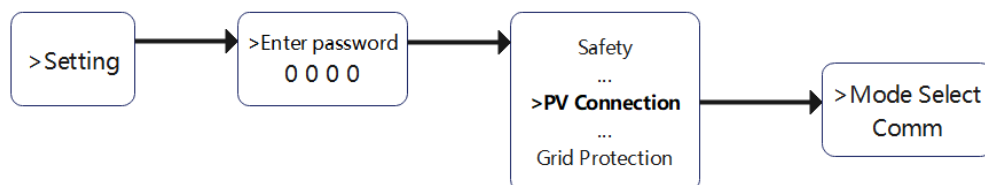
As in the above case, the output current of the optimized PV panel is high and may exceed the input current range of the inverter. However, this problem will be solved if Y-branch connectors are used. This connector will split the current from the PV panel into two parts for each of the two MPPT inputs of the inverter.

The wiring for this application scenario is shown in the diagram below:



Associated configuration

This function can be set up via the LCD on the inverter, see the diagram below (the diagram shows the X3-MIC as an example):



*Please refer to the manual for detailed passwords